Chapter 11: Analysis of Covariance (ANCOVA)

Statistics for Environmental Science

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Chapter Summary **18**

We've now covered regression (Chapter 9) and ANOVA (Chapters 8 and 10). This chapter merges them:

$$\text{Regression} + \text{ANOVA} = \text{ANCOVA}$$

The last chapter compared macroinvertebrate species count across two discrete Pollution categories, present or absent, and two Flow categories, low or high. But suppose you measured each channel's pollution concentration instead of labeling it present or absent.

- Does Flow's effect on species count still depend on pollution once pollution is treated as a continuous measurement instead of two groups? *(covariate: pollution concentration; factor: flow rate)*
- Is the relationship between PFAS exposure and lifetime cancer incidence the same in low-income and high-income American populations? *(covariate: PFAS exposure; factor: income level)*

In each question, a continuous variable is related to the outcome, and a categorical variable splits the observations into groups.

The first question points to a limitation of the last chapter. "Pollution: present or absent" simplifies a continuous measurement, a concentration in parts per billion. Sorting continuous measurements into "high" and "low" buckets throws away the information a covariate needs, which is how much more polluted a channel is, not just whether it is polluted. Whenever you have the measured values, analyze them as a continuous covariate instead of forcing them into groups first. With a categorical factor still in the design, that analysis is ANCOVA. With no categorical factor left, it is plain regression.

Analysis of covariance (ANCOVA) is a linear model with a continuous outcome, at least one continuous predictor (the covariate), and at least one categorical predictor (the factor). ANCOVA is built directly on the regression equation from Chapter 9. A simple linear model is

$$y = \beta_0 + \beta_1 x + \varepsilon$$

Here β_0 is the intercept, β_1 is the slope, and ε is the error. The model describes a single line.

ANCOVA adds a categorical factor to that equation. The chapter asks whether every group follows the same line, whether the groups share a slope but need different intercepts, or whether each group needs its own slope.

We answer it in order: first check that the model explains anything at all, then test whether the groups have different slopes, and only if the slopes are parallel, test whether the groups differ in adjusted means. Step 2 below defines adjusted means, the group means you would expect if both groups had the same average covariate value.

Table 1 places ANCOVA beside regression and the two kinds of ANOVA from Chapters 8 to 10, each written as a model.

Table 1: Four designs as linear models.

Design	Predictors	Model	First question
Regression (Chapter 9)	One continuous	$y = \beta_0 + \beta_1 x + \varepsilon$	Is the slope zero?
One-way ANOVA (Chapter 8)	One categorical factor (three or more groups)	One mean per group	Are all the group means equal?
Factorial ANOVA (Chapter 10)	Two categorical factors, fully crossed	Main effect of A + main effect of B + A × B interaction	Does the effect of one factor depend on the level of the other?
ANCOVA	One categorical factor and one continuous covariate	$y = \beta_0+\beta_1 x+\beta_2 G+\beta_3 xG+\varepsilon$, with $G = 0$ or 1 for the two groups	After the whole-model check, do the groups share a slope?

Why adjust for anything?

Adjusting for a variable means comparing groups at the same value of that variable. **Simpson's paradox**, where a pattern in combined data reverses once you split the data into meaningful groups, shows what can go wrong without adjusting.

The case fatality rate (CFR) is the percentage of confirmed cases that ended in death. During the COVID-19 pandemic, early data showed that Italy's overall case fatality rate was higher than China's. But once researchers split the cases by age group, the trend flipped. In every age group except the youngest, where neither country recorded a death, the fatality rate was higher in China. The apparent contradiction was explained by differences in the age distribution of cases (von Kügelgen et al. 2021).

The numbers below are the ones that study compared. China's cases and deaths by age are as reported on February 17, 2020, and Italy's are as reported on March 9, 2020.

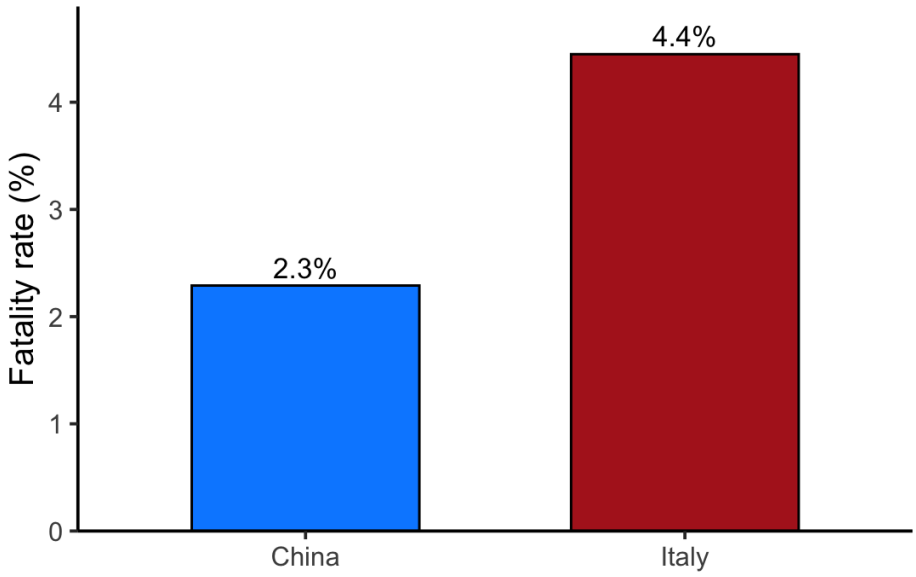


Figure 1: Crude case fatality rates. Italy appears to have a higher fatality rate than China when all cases are combined.

At face value, Italy looks worse. But the two countries' cases might differ in age, and age matters because older people are much more likely to die from COVID than younger people. In these numbers, 13.2% of Italy's cases aged 80 and over died, against none of the 43 cases under 10. So we split cases into 10-year age bands. Within every band where anyone died, China's fatality rate is higher.

Table 2: Fatality rate (%) by age band. China's rate is higher in every band except 0–9, where neither country recorded a death. Several bands under 50 are too small to see in the figure.

Age band	Italy	China
0–9	0.0	0.0
10–19	0.0	0.2
20–29	0.0	0.2
30–39	0.0	0.2
40–49	0.1	0.4
50–59	0.2	1.3
60–69	2.5	3.6

Table 2: Fatality rate (%) by age band. China's rate is higher in every band except 0–9, where neither country recorded a death. Several bands under 50 are too small to see in the figure.

Age band	Italy	China
70–79	6.4	8.0
80+	13.2	14.8

Table 2 lists the rates themselves. The reversal happens because Italy's cases were older on average and China's were younger (Figure 3). A country's overall fatality rate is an average of its age-band rates, weighted by the share of that country's cases in each band. Italy had a larger share of its cases in the oldest bands, where fatality rates are highest, so its overall rate is pulled up. China had a larger share of its cases in the younger bands, where fatality rates are low, so its overall rate is pulled down. Aggregating without adjusting for age hid the within-group pattern entirely. Age is a confounding variable (Section 2.5.6), related both to which country a case came from and to whether the case was fatal.

This kind of reversal happens when the groups differ in their mix of a third variable, here age. Once you compare within each level of the third variable, here within each age band, the within-group pattern shows up again.

ANCOVA handles this situation when the third variable is continuous, instead of grouped into categories like the age bands above. It compares groups at the same covariate value, so a difference between the groups on that covariate does not distort the comparison.

Covariates and fixed factors

A **covariate** is a continuous predictor that is related to the outcome and helps explain part of its variation. In environmental studies, common covariates include temperature, rainfall, soil moisture, and pollution concentration. Including one lets us remove the variation it explains, so the test of the group difference is clearer.

A **fixed factor** is a categorical predictor whose levels the researcher chose on purpose, such as urban vs. rural or shaded vs. open canopy. A **random factor** is a categorical predictor whose levels are a sample from a larger population of possible levels. For example, the counties your streams are in could be a random factor. You aren't interested in those ten counties for their own sake. They represent a larger population of counties, and you want your conclusions to apply beyond the ones you sampled.

Every categorical predictor in this chapter is a fixed factor. The distinction between fixed and random factors matters once you build mixed-effects models, which contain both kinds of factor. This course does not cover them. Mixed-effects models are a different idea from the mixed design (Section 10.9.2), which mixes between-subjects and within-subjects factors.

What ANCOVA is

ANCOVA asks whether a *relationship* holds up across groups. That framing fits environmental work particularly well, because you usually can't experimentally strip out a covariate like temperature, rainfall, or season anyway. Often you wouldn't want to. It may be the reason you designed the study in the first place. In this chapter's main example, the covariate's relationship with the outcome is the question, and the interaction is the result you are testing for.

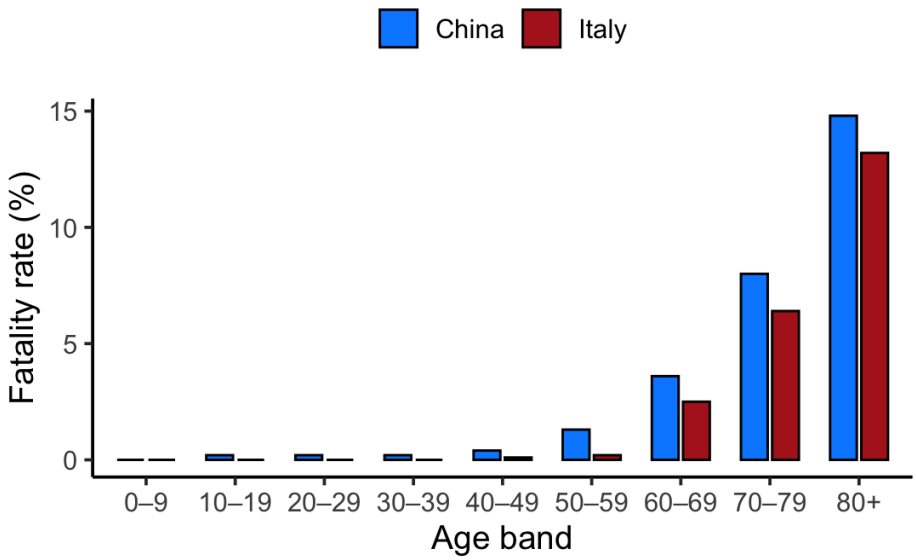


Figure 2: Fatality rates by age group. In every age band where anyone died, China's CFR exceeds Italy's.

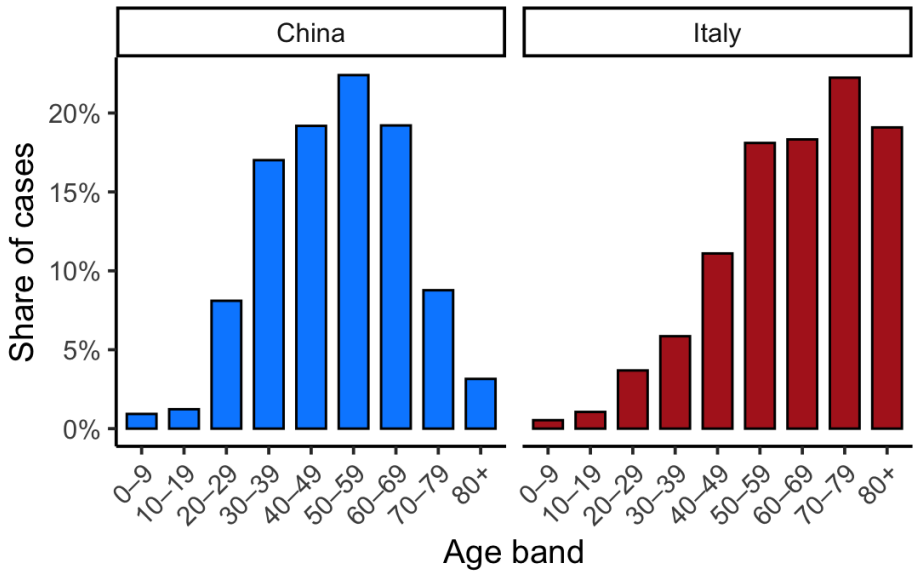


Figure 3: Case age distribution by country. Italy's cases are older on average, and China's are younger.

> **Note: ANCOVA with a nuisance covariate**
>
> Elsewhere the covariate is a nuisance instead, a variable you adjust for so a group comparison is fair (the second example at the end of this chapter). In that version, you test the interaction to check whether the slopes can be treated as parallel before comparing the groups.
> A classic version from psychology compares anxiety scores across two therapy groups after treatment, adjusting for pre-treatment scores.
> Both versions use the same test and the same math. They differ only in which term you came to interpret, so decide that before you fit anything.

For ANCOVA, you need:

1) A continuous dependent variable (the outcome)
2) At least one continuous independent variable (covariate)
3) At least one categorical independent variable (a fixed factor, in this course)

Indicator variables and how they work

Before reading coefficients off an ANCOVA table, it helps to see how one equation can represent two groups at once.

An **indicator variable** is a variable that equals 1 for observations in one group and 0 for all other observations. Categorical predictors get into a linear model as indicator variables (you'll also see them called dummy variables). For a factor with k groups you include $k-1$ indicators. The group left out needs no indicator of its own, because an observation with 0 on every indicator must belong to it. That group is the **reference group**, and every coefficient involving the factor is interpreted relative to it. With two groups, one indicator does the whole job:

$$G = \begin{cases} 0 & \text{group 1 (the reference group)} \\ 1 & \text{group 2} \end{cases}$$

In R, `lm(y ~ x + Group)` creates these indicators for you automatically.

An indicator can change the intercept, the slope, or both. When an indicator variable is turned on (set to 1), it activates the terms multiplied by that variable. Where the indicator appears in the model decides which of these changes.

Consider two predictors: a continuous variable x and an indicator G (0 or 1). The numbers below are arbitrary, chosen to make the arithmetic clean. The worked example later in this chapter plugs in fitted coefficients for air pollution and area type. Use this model:

$$y = 3 + 2x - 7G + 1(xG)$$

When $G = 0$ (indicator off), only the baseline terms are active:

$$y = 2x + 3 \qquad \text{slope} = 2,\ \text{intercept} = 3$$

When $G = 1$ (indicator on), the term $1(xG)$ becomes $1x$ and the term $-7G$ becomes -7. Adding them to the baseline terms and collecting the x terms and the constants gives

$$\begin{aligned} y &= (2x + 3) + (1x - 7) \\ &= (2 + 1)x + (3 - 7) \\ &= 3x - 4 \end{aligned}$$

so the slope is 3 and the intercept is -4.

Turning on the indicator adds $+1$ to the slope and -7 to the intercept. The one equation gives two different lines, and the value of the indicator decides which one applies.

Linear models represent different groups this way. The indicator (G) and its interaction with x let a single equation shift the intercept and the slope for each group. ANCOVA uses exactly this logic to test, in order, whether the slopes differ between groups (the xG term) and, if they don't, whether the adjusted means differ (the G term).

Naming the pieces. Written in the general form, the same equation gives each coefficient a name and a meaning:

$$y = \underbrace{\beta_0}_{\text{reference intercept}} + \underbrace{\beta_1 x}_{\text{reference slope}} + \underbrace{\beta_2 G}_{\text{intercept shift}} + \underbrace{\beta_3 xG}_{\text{slope shift}} + \varepsilon$$

- β_0 and β_1 describe the line for the reference group ($G = 0$).
- β_2 is how much the intercept moves for the other group.
- β_3 is how much the slope changes for the other group. This is the interaction term. In Chapter 10, an interaction meant that the effect of one factor depended on the level of the other factor. Here it means that the effect of x, its slope, depends on the group. Both are the same idea, because an interaction means the effect of one predictor depends on the value of another, whether that predictor is a factor or a covariate.

The **parallel-lines model** includes β_2 but leaves out β_3, so the intercept can shift between groups but the slope cannot, and the two lines are parallel. When the slope difference is not significant, we use this simpler model instead of keeping the interaction term.

> **i** Note: How R names the four coefficients
>
> These are the four coefficients R reports for this model, and each one can be checked against a graph of the two group lines. For `lm(y ~ x * Group)`, where `Group` has levels A (the reference) and B, R names the rows `(Intercept)` for β_0, `x` for β_1, `GroupB` for β_2, and `x:GroupB` for β_3. The worked example below follows the same pattern: `(Intercept)`, `AirPollution`, `AreaTypeUrban`, and `AirPollution:AreaTypeUrban`.

The steps of an ANCOVA

Model. For one covariate (x) and a two-level group indicator $G \in \{0, 1\}$,

$$y = \beta_0 + \beta_1 x + \beta_2 G + \beta_3 (x \times G) + \varepsilon$$

Figure 4 shows the three possibilities this model can produce.

ANCOVA works right to left across that figure. It first asks whether the rightmost panel is what we have (do the slopes differ?), and only if the answer is no does it ask whether we have the middle panel or the leftmost (are the parallel lines at different heights?).

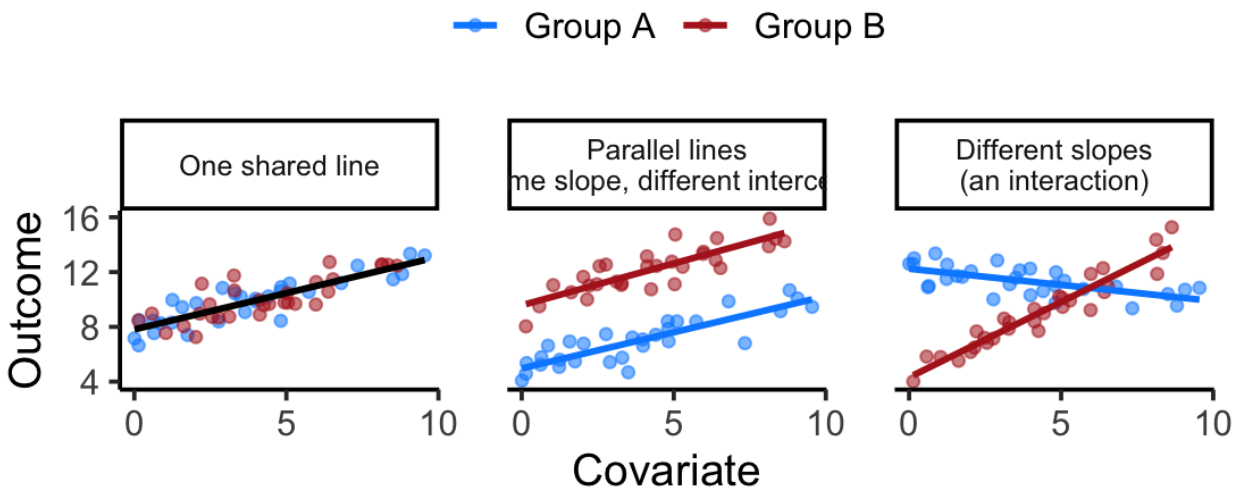


Figure 4: The question ANCOVA answers. Left: one line describes both groups. Middle: the groups need different intercepts but share a slope, so the lines are parallel. Right: the groups need different slopes, so the lines are not parallel and the gap between them depends on where you look.

Step 0: Does the model explain anything? Before looking at any single term, check the whole-model F-test, the regression F from Section 9.4 applied to all the terms at once, which asks whether the covariate, the group, and their interaction together explain more of y than chance would.

- **Null (H_0):** none of the three terms explains y, $\beta_1 = \beta_2 = \beta_3 = 0$.
- **Alternative (H_a):** at least one of β_1, β_2, β_3 is not 0.

R reports it on the last line of `summary(lm(...))`. If it is not significant, stop. The data give no evidence that the covariate, the group, or their interaction explains y, so no single term is worth interpreting.

Step 1: Are slopes equal across groups?

- **Null ($H_{0,\mathbf{int}}$):** parallel lines, the same slope in both groups, $\beta_3 = 0$.
- **Alternative ($H_{a,\mathbf{int}}$):** different slopes, $\beta_3 \neq 0$.

If Step 1 rejects. The slopes differ, so stop testing the group term. There is no single group difference to test, because the gap between the lines changes along the covariate. Describe each group's line instead. The model gives both slopes (the reference group's is β_1, the other group's is $\beta_1 + \beta_3$), but it does not test whether each one differs from zero. To test each slope, fit a separate simple regression within each group.

Step 2: Are adjusted means equal? *(only if Step 1 did not reject $H_{0,int}$, so the slopes are treated as parallel)* Drop the interaction and refit the parallel-lines model,

$$y = \beta_0 + \beta_1 x + \beta_2 G + \varepsilon$$

- **Null ($H_{0,\mathbf{group}}$):** no difference between the groups at the same covariate value, $\beta_2 = 0$.
- **Alternative ($H_{a,\mathbf{group}}$):** the groups differ at the same covariate value, $\beta_2 \neq 0$.

With parallel lines, β_2 is the vertical gap between the two lines, and it is the same at every value of x. The **adjusted difference** is how much the groups differ when they are compared at the same covariate value, so with parallel lines it equals β_2. The **adjusted means** are the group means you would expect if both groups had the same average covariate value. To get them, evaluate each group's line at the overall mean of x. Raw group means compare the groups at whatever covariate values each group happens to have. Adjusted

means compare them at one shared covariate value, the overall mean of x. So if one group happens to have higher covariate values, that difference does not enter the comparison. The second example at the end of this chapter works through one.

In the parallel-lines model, β_1 is the common slope.

Intercepts and centering. Intercepts are evaluated at $x = 0$. In the model with an interaction, the lines can have different slopes, so β_2 compares the groups at $x = 0$ only, which may be far outside the data. **Centering** the covariate means subtracting its mean from every value, $\tilde{x} = x - \bar{x}$. After centering, $\tilde{x} = 0$ where $x = \bar{x}$, so the intercept and β_2 now describe the lines at the average covariate value. In the parallel-lines model the gap is the same everywhere, so centering changes the intercept but not β_2.

What to report.

- If $x \times G$ is significant: "The effect of x differs by group," then each group's line with its own slope test, and a figure with separate fitted lines.
- If $x \times G$ is not significant: the common slope β_1 and the adjusted group difference β_2 (or the adjusted means), with a figure showing parallel lines.

As always, statistical significance doesn't guarantee ecological or practical importance. Interpret a rejected null in light of the study's design, measurement precision, and the actual size of the effect, not just its p-value.

> Warning: Two ways to misread an ANCOVA
>
> - Interpreting the group term when the interaction is significant. With different slopes, the gap between the groups changes along the covariate, so no single group difference or adjusted mean describes it.
> - Comparing intercepts when x is not centered. The intercepts then compare the groups at $x = 0$, which may be outside the data.

Checking ANCOVA's assumptions

Most of these are ones ANOVA already needed. Three are new because a covariate has entered the model (Table 3).

Table 3: Assumptions of ANCOVA and how to check them.

Assumption	New with ANCOVA?	How to check
Independent observations	No (universal, Section 6.3.5)	The study design
Normal residuals and similar variances across groups	No (Sections [-8.5.2] and [-9.6])	Q-Q plot, residuals-versus-fitted plot, Levene's test
Linearity within each group	Yes	A scatter plot with a fitted line for each group
Homogeneity of slopes (the groups' lines are parallel)	Yes	The interaction test (Step 1), not a check run beforehand
The covariate is measured with little error and is not affected by the group factor (the treatment)	Yes	The study design

If the covariate is measured with error, ANCOVA only partly adjusts for group differences in the covariate, and the leftover difference can look like a treatment effect. If the treatment changes the covariate, adjusting for the covariate removes part of the treatment's own effect, and the adjusted difference understates it.

Table 4 collects the three steps of an ANCOVA on one card.

Table 4: Test card: ANCOVA.

Purpose	Does the covariate's relationship with the outcome differ between groups? If not, do the groups differ at the same covariate value?
Hypotheses	Step 0: $H_0 : \beta_1 = \beta_2 = \beta_3 = 0$ Step 1: $H_{0,\text{int}} : \beta_3 = 0$ (parallel lines) Step 2, only if Step 1 fails to reject: $H_{0,\text{group}} : \beta_2 = 0$ in the parallel-lines model
Assumptions	Normal residuals and similar variances across groups Linearity within each group Homogeneity of slopes (tested in Step 1) A covariate measured with little error and not affected by the treatment
Test statistic	Model: $y = \beta_0 + \beta_1 x + \beta_2 G + \beta_3 (x \times G) + \varepsilon$ Step 0: whole-model $F(3, N-4)$ Step 1: $F(1, N-4)$ for the $x \times G$ row Step 2: $t = b_2 / SE(b_2)$, $df = N - 3$
Report	If $x \times G$ is significant: its $F(df)$, p, model R^2, and each group's slope from a separate regression with $t(df)$ and p If not: the adjusted difference b_2 with $t(df)$, p and 95% CI, the adjusted means, the common slope, and the interaction F

Full ANCOVA example

Birds, air pollution and area type

Suppose we are studying the effect of air pollution, measured as PM2.5 (fine particulate matter, in µg/m³), on bird density, in birds per hectare, and we want to know whether that relationship differs between urban and rural areas. Figure 5 lays out the design.

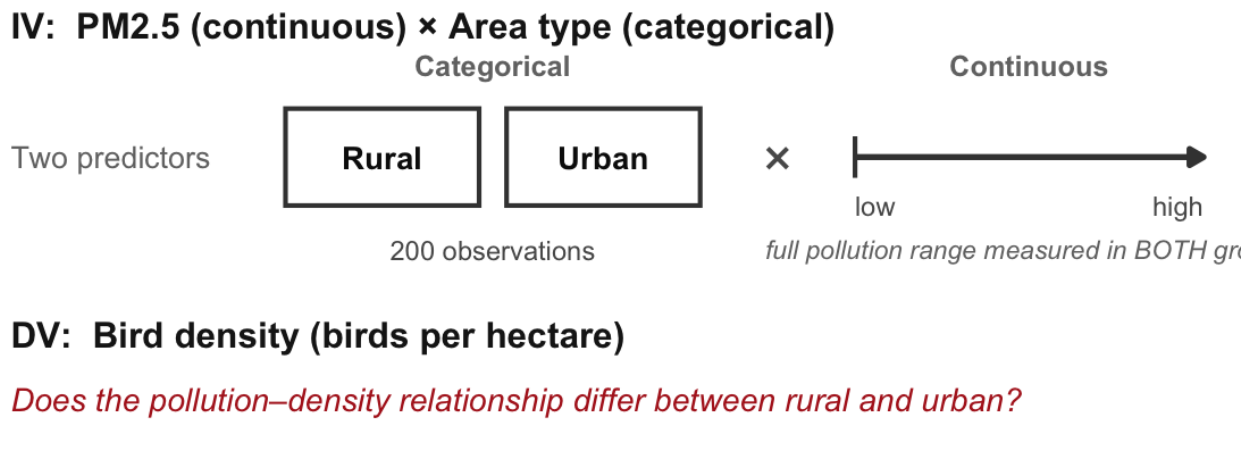


Figure 5: An ANCOVA design: one continuous predictor and one categorical factor.

This design has the same structure as the factorial design in Chapter 10. Both cross two predictors and ask whether one changes the effect of the other. The only difference is that factorial's second predictor has levels, while ANCOVA's is a continuous range.

Both groups span the full pollution range, so rural sites are not simply the clean ones. An adjusted comparison compares the groups at the same pollution level, and that only works where both groups have sites at that level. Without the overlap, each group's line would have to be extended past its own data, the extrapolation problem from Section 9.7.

We might expect pollution to have a stronger negative effect in urban areas. Urban habitat is already fragmented and offers birds fewer clean patches to retreat to, while rural areas have more intact habitat nearby.

In the ANCOVA model, area type enters as an indicator variable, and air pollution enters as the covariate.

Null and alternative hypotheses in our example ANCOVA

First, we state the hypotheses for each step:

0. **Whole model.** Do air pollution, area type, and their interaction together explain bird density?

   $H_0 : \beta_1 = \beta_2 = \beta_3 = 0$

1. **Slopes (interaction).** Does air pollution affect bird density the same way at urban and rural sites?

   $H_{0,\text{int}} : \beta_3 = 0$ (parallel lines)

   $H_{a,\text{int}} : \beta_3 \neq 0$

   A significant interaction means the slopes differ, and air pollution's relationship with bird density depends on area type.

2. **Adjusted means (only if the slopes are parallel).** At the same level of air pollution, do urban and rural sites differ in bird density?

   $H_{0,\text{group}} : \beta_2 = 0$

   $H_{a,\text{group}} : \beta_2 \neq 0$

The data

The data in this example are invented. There are 200 sites, about half urban and half rural, with PM2.5 between 5 and 30 µg/m³. The true slope is −0.25 birds per hectare for each 1 µg/m³ of PM2.5 at rural sites and −0.75 at urban sites, plus random scatter, so the fitted slopes below come out near these values but not on them. In this draw the fitted urban slope is about twice the rural slope, though the true urban slope is three times as steep.

Before fitting the model, look at the data. Figure 6 shows the outcome, bird density, by area type, which gives a quick sense of whether urban and rural areas differ on average.

Bird density is higher at rural sites, with a median of 35 birds against 25 at urban sites, consistent with what we might expect ecologically.

Next, check the covariate, air pollution. If the covariate differs sharply between groups, ANCOVA adjusts for that difference. If it doesn't, the adjusted group difference will be close to the raw difference in group means.

Air pollution is similar at urban and rural sites. The medians are 17.1 at rural sites and 15.2 at urban sites, a small gap compared with the range of about 5 to 30 µg/m³. This plot doesn't test anything, but it tells us

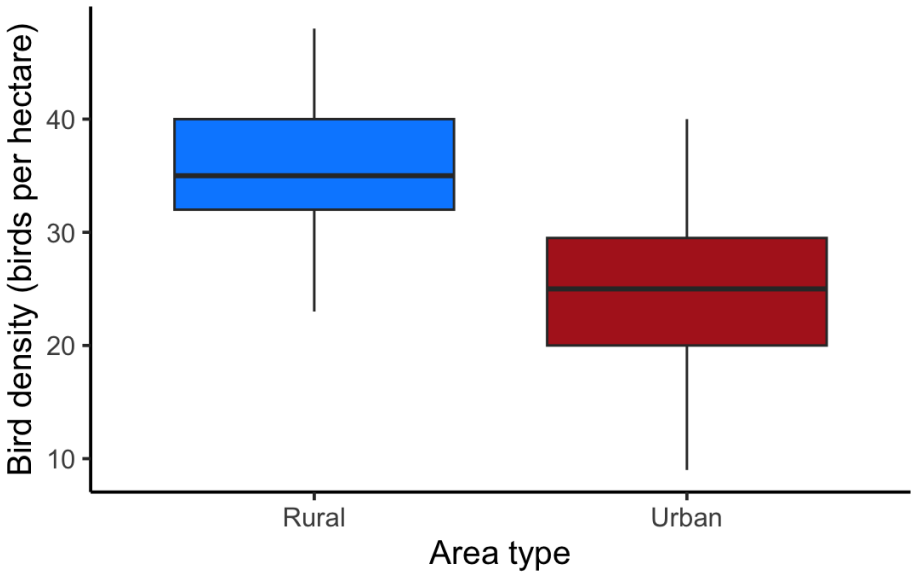


Figure 6: Bird density by area type. Rural sites have higher bird density than urban ones.

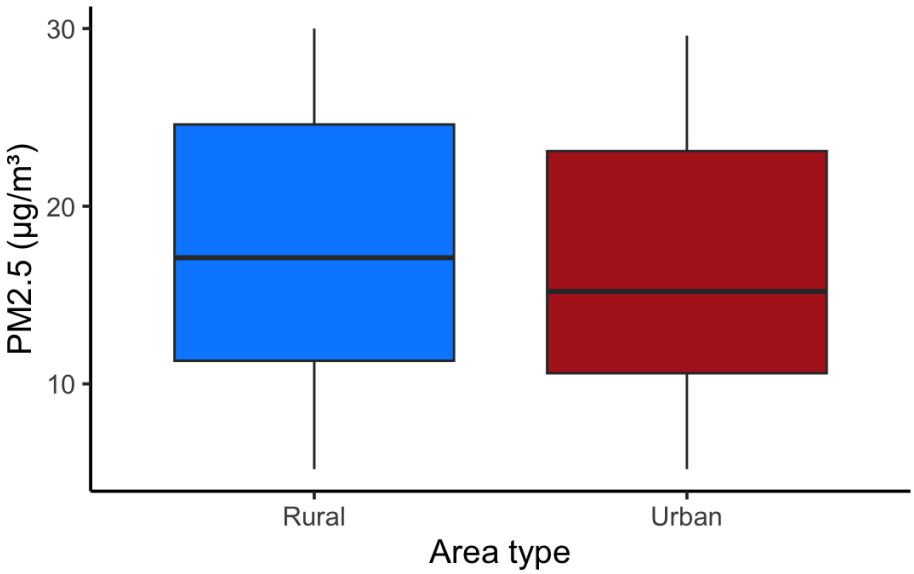


Figure 7: Air pollution by area type. The two groups look broadly similar, so there is no large covariate imbalance for the model to adjust away.

what to expect. Because pollution is similar in the two groups, a group difference in bird density is unlikely to come from a difference in pollution.

Results of the ANCOVA model

Fit the model. This is the same `aov()` call we used for one-way and factorial ANOVA. Because `AirPollution` is numeric instead of a factor, `aov()` treats it as a continuous covariate, so R needs no separate ANCOVA function.

```
ancova <- aov(BirdDensity ~ AirPollution * AreaType, data = data)
```

Table 5: ANCOVA table for bird density

Source	df	SS	MS	F	p
AirPollution	1	2190.91	2190.91	88.73	< 0.001
AreaType	1	6459.57	6459.57	261.59	< 0.001
AirPollution:AreaType	1	271.34	271.34	10.99	0.001
Residuals	196	4839.85	24.69		

Step 0: the whole model. `summary(lm())` reports $F(3, 196) = 120.44$, $p < 0.001$, with $R^2 = 0.65$. The model explains more of the variation in bird density than chance would, so we continue.

Step 1: the interaction. The `AirPollution:AreaType` row tests $H_{0,\text{int}}$: $F(1, 196) = 10.99$, $p = 0.001$. We reject $H_{0,\text{int}}$, so the slopes are not parallel. Air pollution's relationship with bird density depends on area type, so we stop here and do not test adjusted means.

What the other two rows are. The table still has rows for `AirPollution` and `AreaType`, both with $p < 0.001$, so it is tempting to read them, but they don't answer this chapter's question. R builds these rows in sequence, which means each row tests its term after the terms listed above it have been fitted. `AirPollution` tests one average slope for all 200 sites, ignoring area type, and `AreaType` tests a group difference after that single slope is fitted. Once the interaction says the slopes differ, neither row describes the data. There is no single slope, and the group difference depends on where along the pollution gradient you look. Because the rows are built in sequence and the groups are unequal in size (97 rural, 103 urban), listing the terms in a different order would also change their sums of squares.

Reading the coefficients.

Fitting the same model with `lm()` shows the regression coefficients directly, so we can see how each term contributes to predicting bird density.

```
ancova_lm <- lm(BirdDensity ~ AirPollution * AreaType, data = data)
```

Table 6: Linear model coefficients

Term	Estimate	SE	t	p
(Intercept)	41.780	1.310	31.88	< 0.001
AirPollution	-0.346	0.068	-5.07	< 0.001
AreaTypeUrban	-5.895	1.803	-3.27	0.001
AirPollution:AreaTypeUrban	-0.319	0.096	-3.31	0.001

R automatically converts the categorical `AreaType` variable into a 0/1 indicator, the same G from the indicator-variable section earlier in this chapter. It picks one level (here, "Rural," alphabetically first) as the reference, and every coefficient involving `AreaType` is interpreted relative to that reference. That's why you see a coefficient called `AreaTypeUrban` but no separate `AreaTypeRural` term.

These coefficients give the *direction* and *magnitude* of each effect:

- **`(Intercept)`.** Estimated mean bird density for rural sites (the reference group) when air pollution is zero.
- **`AirPollution`.** Slope of the relationship between pollution and bird density in rural sites. This is not the same quantity as the `AirPollution` row of the ANOVA table above, which was one average slope for both groups.
- **`AreaTypeUrban`.** Difference in intercept between urban and rural areas, which compares the groups at air pollution = 0.
- **`AirPollution:AreaTypeUrban`.** Difference in slope between urban and rural areas (the interaction).

Visualize the interaction.

A scatter plot with fitted regression lines (Figure 8) lets us see how air pollution relates to bird density for each area type.

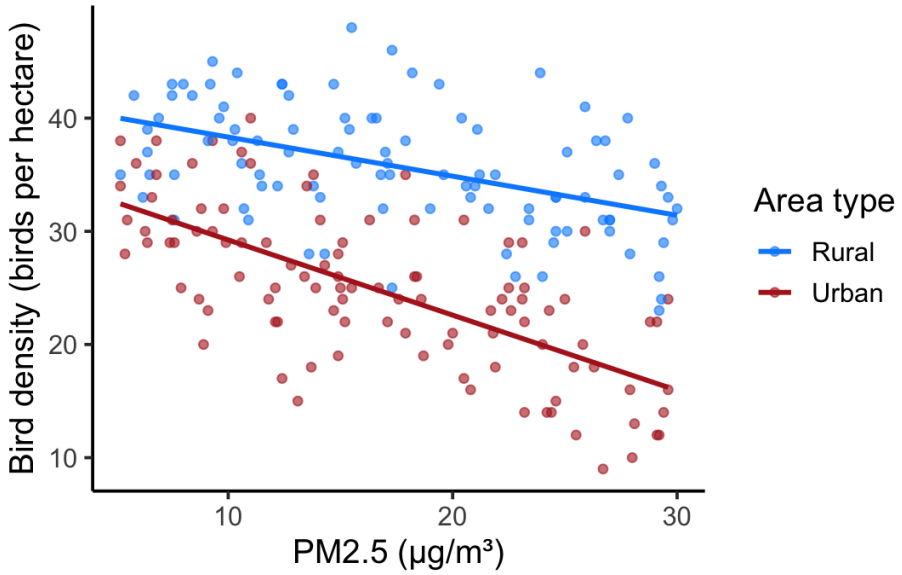


Figure 8: Bird density against air pollution, with a separate fitted line for each area type. Both lines slope down, but the urban line falls about twice as steeply.

The two fitted lines have different slopes, which is the interaction the test detected. Both lines slope downward, so pollution is associated with fewer birds in both settings, but the urban line drops roughly twice as fast. The lines are about 7.6 birds apart at the lowest pollution levels and about 15.5 apart at the highest.

Testing each group's slope. The interaction told us the two slopes differ from each other. It did not tell us whether each one differs from zero, so we fit a separate simple regression within each group.

```
lm(BirdDensity ~ AirPollution, data = subset(data, AreaType == "Rural"))
lm(BirdDensity ~ AirPollution, data = subset(data, AreaType == "Urban"))
```

Table 7: Separate regressions of bird density on air pollution within each area type.

Area type	Slope	SE	t	df	p
Rural	-0.346	0.065	-5.29	95	< 0.001
Urban	-0.665	0.070	-9.43	101	< 0.001

Both slopes are significant. Bird density falls with pollution at rural sites and falls about twice as steeply at urban sites. The slope estimates match the ANCOVA's, -0.346 for rural and $-0.346 + (-0.319) \approx -0.665$ for urban, because a model with an interaction fits each group its own line. Only the tests are new, and their standard errors differ slightly because each regression estimates its own error variance.

The points scatter around each line. The model's R^2 is 0.65, so air pollution, area type, and their interaction explain about 65% of the variation in bird density. The remaining 35% is the random scatter built into these invented data. In field data, that remainder would include other factors, measurement noise, and natural variability. An R^2 this large is strong by the standards of field data, and it is large here partly because the data were invented to illustrate the method.

Centering the covariate.

Interpreting intercepts at $x = 0$ can be misleading when "zero pollution" isn't meaningful.

By centering the covariate (subtracting its mean), we make the intercept the predicted bird density for rural sites at mean air pollution instead.

Table 8: Coefficients with the covariate centered

Term	Estimate	p
(Intercept)	35.828	< 0.001
AirPollution_c	-0.346	< 0.001
AreaTypeUrban	-11.393	< 0.001
AirPollution_c:AreaTypeUrban	-0.319	0.001

Centering doesn't change the slopes, the interaction, or their p-values. It moves the intercept and the group term, and the group term's test now compares the groups at average pollution instead of at zero, so its p-value changes too.

Interpreting the centered model. The mean pollution level is about 17 µg/m³.

- **Intercept (35.8).** Predicted bird density for rural sites at average air pollution.
- **`AreaTypeUrban` (-11.4).** At the mean pollution level, urban sites have about 11 fewer birds than rural sites.
- **`AirPollution_c` (-0.346).** Slope for rural sites: bird density falls by about 0.35 birds per hectare for each additional 1 µg/m³ of PM2.5.
- **Interaction (-0.319).** Difference in slopes between groups. The urban slope $= -0.346 + (-0.319) \approx -0.66$, roughly twice as steep a decline as in rural sites.

The group difference changed when we centered, from -5.9 to -11.4. Uncentered, the term compares groups at pollution = 0, a value outside the observed data. Centered, it compares them at average pollution, a value inside the observed range. When slopes differ, "the group difference" is not one number. It depends on where along the covariate you look.

Summarize and interpret.

The significant interaction shows that air pollution's effect on bird density depends on area type. Bird density declines with pollution in both settings, but the decline is about twice as steep in urban sites. Because the interaction is significant, we focus on these different slopes, not on adjusted mean differences.

If the interaction had *not* been significant, we would instead drop the interaction term and interpret the parallel-lines model, comparing adjusted means between groups. The next example shows that case.

! Results reporting: ANCOVA

> Results from the ANCOVA indicated that air pollution's effect on bird density depended on area type, $F(1, 196) = 10.99$, $p = 0.001$, model $R^2 = 0.65$. Separate regressions showed that bird density declined with increasing pollution at both rural sites, $b_1 = -0.35$ birds per hectare for each 1 µg/m³ of PM2.5, $t(95) = -5.29$, $p < 0.001$, and urban sites, $b_1 = -0.66$, $t(101) = -9.43$, $p < 0.001$, with the urban decline roughly twice as steep. These findings suggest that urbanization intensifies the pollution-bird density relationship and may point toward the need for area-specific conservation strategies.

Second example: when the covariate is a nuisance

In the bird example, the covariate relationship was the question, and the interaction was the finding. The other way you'll see ANCOVA reverses those roles. A **nuisance covariate** is a covariate you adjust for so that a group comparison is fair, when the group comparison is the question. This is also the continuous version of the Simpson's paradox problem the chapter opened with.

Table 9 compares the two ways this chapter uses ANCOVA.

Table 9: Two roles for the covariate in ANCOVA.

Role of the covariate	The covariate is	What you interpret	Example in this chapter
Covariate as the question	The relationship you came to study	The interaction term β_3, which tests whether the slopes differ	Air pollution and bird density, urban vs. rural sites
Nuisance covariate	Something to adjust for so the group comparison is fair	The group term β_2, the adjusted difference, from the parallel-lines model	Occupancy and electricity use, retrofitted vs. standard buildings

Retrofits and electricity use

A city facilities office wants to know whether energy retrofits (new insulation, lighting, and controls) reduced electricity use. It compares 30 retrofitted buildings with 30 standard ones. Electricity use depends heavily on how many people are in a building, and the retrofitted buildings happen to be the busier ones. These buildings are invented. In the invented data, a retrofit cuts electricity use by 40 kWh per day at any occupancy, both kinds of building use 2 more kWh per day for each extra occupant, and retrofitted buildings hold 60 to 120 people against 30 to 90 for standard ones.

Occupancy also passes the last check on the assumptions list. A retrofit changes a building's equipment, not how many people use it, so adjusting for occupancy doesn't remove any of the retrofit's own effect.

The raw comparison

Average daily occupancy is 93.5 people in the retrofitted buildings and 59.4 in the standard ones. Figure 9 compares electricity use without adjusting for occupancy. Retrofitted buildings use more, 342.1 vs. 318.0 kWh per day, and a two-sample t-test calls that difference significant, $t(58) = 2.16$, $p = 0.035$. Taken at face value, the retrofits made things worse.

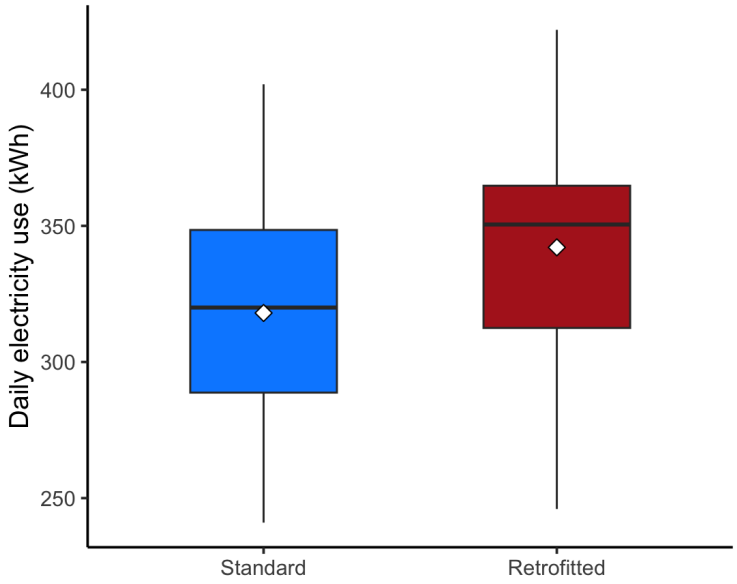


Figure 9: Daily electricity use by building type, ignoring occupancy. Diamonds mark the group means.

The ANCOVA

Step 0: the whole model. $F(3, 56) = 56.61$, $p < 0.001$. The model explains more of the variation in electricity use than chance would, so we continue.

Step 1: the interaction. The `Occupancy:Type` row gives $F(1, 56) = 0.01$, $p = 0.927$. We fail to reject $H_{0,\text{int}}$, so we treat the slopes as parallel. Each additional occupant adds about the same amount of electricity in either kind of building. We drop the interaction and refit.

Step 2: the adjusted difference. Table 10 shows the parallel-lines model.

```
lm(Energy ~ Occupancy + Type, data = energy)
```

Table 10: Parallel-lines model: electricity use on occupancy and building type.

Term	Estimate	SE	t	p
(Intercept)	202.23	10.15	19.92	< 0.001
Occupancy	1.95	0.16	12.48	< 0.001
TypeRetrofitted	-42.46	7.90	-5.38	< 0.001

The `Occupancy` coefficient is the common slope, about 1.95 kWh per day for each additional occupant. The `TypeRetrofitted` coefficient is b_2, the estimate of β_2, the adjusted difference. At the same occupancy,

retrofitted buildings use 42.5 kWh per day less than standard ones, $t(57) = -5.38$, $p < 0.001$. The t has 57 degrees of freedom, 60 buildings minus the 3 coefficients the model estimates. We reject $H_{0,\text{group}}$.

Figure 10 shows what that coefficient measures. The adjusted means express the same comparison as one value for each building type. Evaluating each line at the overall mean occupancy, 76.4 people, gives 351.3 kWh per day for standard buildings and 308.8 for retrofitted ones. Their difference is b_2.

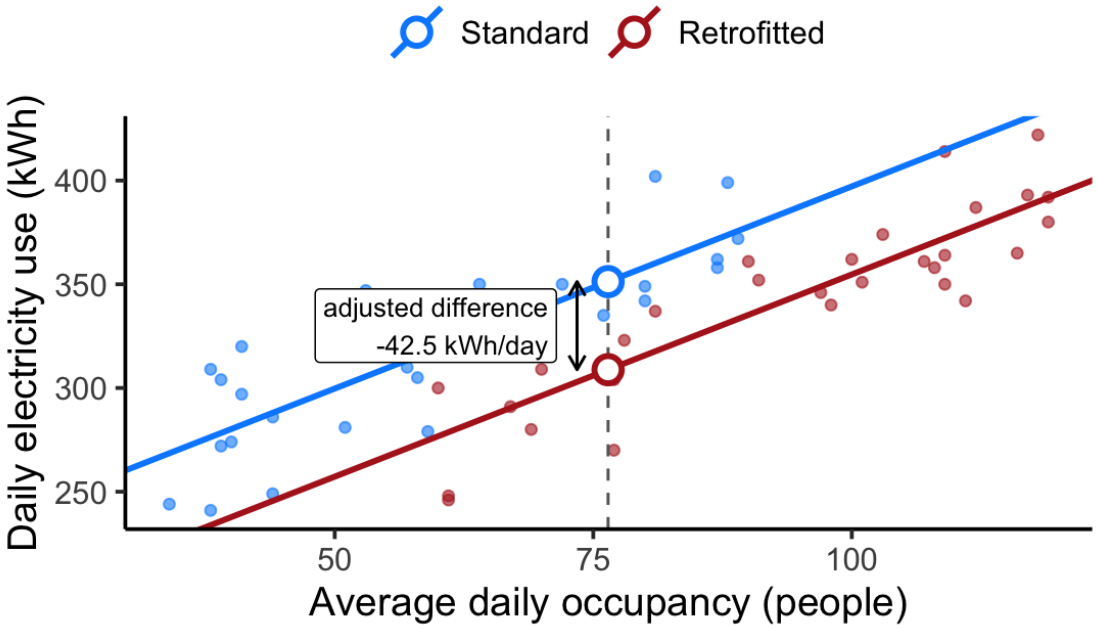


Figure 10: Electricity use against occupancy, with the parallel lines from the model. Open circles mark the adjusted means, each line's value at the overall mean occupancy (dashed line). The arrow is the adjusted difference, which is the same at every occupancy.

The raw comparison and the adjusted one point in opposite directions, for the same reason as in the COVID example. The groups differed in a third variable. The retrofitted buildings were busier, and busier buildings use more electricity. Compared at the same occupancy, retrofitted buildings use less electricity.

Adjusting for occupancy also shrank the error term. In the raw comparison the error mean square is 1864, and in the parallel-lines model it is 508. Much of the building-to-building variation in electricity use comes from occupancy, and once the model accounts for it, that variation no longer counts as error. Repeated-measures ANOVA gains power the same way in Section 8.6. There the known source of variation was the units themselves (the monitoring stations), and here it is a measured covariate. Either way, removing it from MS_{Error} makes the denominator of the test smaller, so a real group difference is easier to detect.

! Results reporting: ANCOVA with parallel slopes

> After adjusting for occupancy, retrofitted buildings used less electricity than standard buildings, $b_2 = -42.5$ kWh per day, $t(57) = -5.38$, $p < 0.001$, 95% CI [-58.3, -26.7] kWh per day. Adjusted means at the average occupancy of 76.4 people were 308.8 kWh per day for retrofitted buildings and 351.3 for standard ones. The occupancy × building type interaction was not significant, $F(1, 56) = 0.01$, $p = 0.927$, so a common slope was used (1.95 kWh per day per occupant).

Chapter Summary

Why it matters. Environmental relationships rarely play out identically everywhere. ANCOVA tests whether a continuous relationship, such as pollution's effect on bird density, is the same in every group. When it is, ANCOVA also compares the groups at the same covariate value.

Core ideas

- **ANCOVA is regression plus a categorical factor.** It takes the familiar $y = \beta_0 + \beta_1 x + \varepsilon$ line and lets a categorical grouping shift the intercept, the slope, or both, using the same indicator-variable machinery that underlies every linear model in this book.
- **Check the whole model, then the interaction.** If the whole-model F is not significant, stop. Otherwise test whether the groups' slopes differ ($H_{0,\text{int}} : \beta_3 = 0$). That answer decides how everything else gets interpreted.
- **If the interaction is significant, report the two lines.** If slopes differ, describe how the relationship changes by group instead of reducing it to a single adjusted-mean comparison, which assumes parallel lines. Test each group's slope with its own regression, since the interaction only says the slopes differ from each other.
- **If slopes are parallel, compare adjusted means.** Drop the interaction term and test β_2, the gap between the parallel lines. When the groups differ on the covariate, the adjusted comparison can point the opposite way from the raw one, as it did for the retrofitted buildings.
- **A covariate also shrinks the error term.** Variation the covariate explains leaves the error mean square, which is the same way repeated-measures ANOVA gains power by removing unit-to-unit variation.
- **Center the covariate when the intercept should mean something.** Left uncentered, the intercept describes the groups at covariate = 0, a value the data may never actually contain.

von Kügelgen, Julius, Luigi Gresele, and Bernhard Schölkopf. 2021. "Simpson's Paradox in COVID-19 Case Fatality Rates: A Mediation Analysis of Age-Related Causal Effects." *IEEE Transactions on Artificial Intelligence* 2 (1): 18–27. https://doi.org/10.1109/TAI.2021.3073088.